

Day 3Combinational Assignment

- 1) Implement the following functions using
 i) 2:1 mux and NOT gates ii) 4:1 Mux.

a) $F(A, B, C) = \bar{A}C + B\bar{C}$ ~~b)~~

Choose C as select line

For 2:1 Mux

$$F = \bar{S}I_0 + SI_1$$

when $C=0$, $F=B$

when $C=1$, $F=\bar{A}$

Hence, $S=C$, $I_0=B$, $I_1=\bar{A}$

$$F = \text{MUX}(C, B, \bar{A})$$

- b) Using 4:1 Mux, Take A, B as select lines

$A \quad B \quad F$

$$I_1 = A \quad I_0 = B$$

0 0 C

$$I_0 = C \quad I_1 = 1 \quad I_2 = 0 \quad I_3 = \bar{C}$$

0 1 1

1 0 0

$$F = \text{MUX}(A, B, C, 1, 0, \bar{C})$$

1 1 $\bar{C}$

- 2) Implement the functionality of a full adder using 2:1 MUX only

$$S = A \oplus B \oplus C_{in}$$

$$C_{out} = AB + BC_{in} + AC_{in}$$

Using 2:1 mux with select A

$$A \oplus B = \bar{A}B + A\bar{B}$$

Select A.

$$I_0 = B$$

$$I_1 = \bar{B}$$

$$\underline{\underline{X = A \oplus B}}$$

$$S = (A \oplus B) \oplus C_{in}$$

$$X = A \oplus B$$

$$S = X \oplus C_{in}$$

Using 2:1 mux

$X \rightarrow$ select line

$$S = \bar{X}(C_{in}) + X(\bar{C}_{in})$$

$$S = X$$

$$I_0 = C_{in} \quad I_1 = \bar{C}_{in}$$

$$S = (A \oplus B) \oplus C_{in}$$

Carry:

$$C_{out} = AB + AC_{in} + BC_{in}$$

$$F = \bar{S}I_0 + SI_1$$

$$X = A \oplus B$$

$$X = 0$$

$$A = B$$

A	B
0	0
1	1

$$X = 1$$

$$A \neq B$$

A	B
0	1
1	0

when $A = B = 0$

$$C_{out} = 0$$

when $A = B = 1$, $C_{out} = 1$

whenever, $X = 0$

$$C_{out} = A$$

when, $A = 0$, $B = 1$

$$C_{out} = 0 + 0 \cdot C_{in} + 1 \cdot C_{in} = C_{in}$$

when, $A = 1$, $B = 0$

$$C_{out} = 0 + 1 \cdot C_{in} + 0 = C_{in}$$

$$\underline{\underline{C_{out} = C_{in}}}$$

$$y_2 = \Sigma I_0 + \Sigma I_1$$

$$S = X = A \oplus B$$

$$I_0 = A \quad \text{Count} = \bar{X}A + X\bar{C}_{in}$$

$$X = 0 \Rightarrow A$$

$$I_1 = C_{in} \quad \text{Count} = (\bar{A} \oplus B)A + (A \oplus B)C_{in}$$

$$X = 1 \Rightarrow C_{in}$$

3) Simplify $f(a, b, c, d) = \Sigma m(2, 4, 5, 6, 10) + D(12, 13, 14, 15)$ and implement the SOP expression using 4:1 mux.

$ab \backslash cd$	$\bar{c}\bar{d}$	$\bar{c}d$	$c\bar{d}$	cd
$\bar{a}\bar{b}$	0	1	3	2
$\bar{a}b$	4	5	7	6
$a\bar{b}$	12	13	15	14
ab	8	9	11	10

$$\underline{\underline{b\bar{c} + c\bar{d}}}$$

Implement using 4:1 MUX

choose B and C as select lines

$$I_1 = B \quad I_0 = C$$

$$BC = 00, \text{ then } F = 0 \Rightarrow I_0 = 0$$

$$BC = 01, \text{ then } F = D' \Rightarrow I_1 = D'$$

$$BC = 10, \text{ then } F = 1 \Rightarrow I_2 = 1$$

$$BC = 11, \text{ then } F = D' \Rightarrow I_3 = D'$$

$$\underline{\underline{F = B\bar{C} + CD'}}$$